Vijay Janapa Reddi, Ph.D.

Gordon McKay Professor of Electrical & Computer Engineering
Harvard John A. Paulson School of Engineering and Applied Sciences

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Gordon McKay Professor of Electrical & Computer Engineering at Harvard SEAS, Vice President of ML-Commons, and (from July 2026) Visiting Professor at ETH Zürich. Co-architect of the MLPerf Inference benchmark (the industry-standard ML benchmarking suite adopted by Google, Microsoft, NVIDIA, Meta, AMD, and Intel) and founder of the open-source *Machine Learning Systems* curriculum used at 50+ universities across 5 continents, with 100,000+ learners reached through the TinyML edX certificate. Two MIT Press textbooks forthcoming (2026 and 2027). Ph.D., Harvard 2010; previously Associate Professor at UT Austin.

Google Scholar: 21,698 citations · h-index 62 · i10-index 162.

Current Position

Gordon McKay Professor of Electrical & Computer Engineering
John A. Paulson School of Engineering and Applied Sciences
Harvard University

Primary Research Areas

Computer Architecture; Machine Learning Systems; Autonomous Agents/Robotics

Education

2010 Ph.D. in Computer Science, Harvard University
2006 M.S. in Computer Engineering, University of Colorado Boulder
2003 B.S. in Computer Engineering, Santa Clara University

Current and Previous Academic Positions

2026/– Visiting Professor, ETH Zürich (starting July 2026)
2025/– Gordon McKay Professor of Electrical & Computer Engineering, Harvard University
2019–2024 John L. Loeb Associate Professor of Engineering and Applied Sciences, Harvard University
2018/– Adjunct Associate Professor, The University of Texas at Austin
2017–2018 Associate Professor, The University of Texas at Austin
2011–2017 Assistant Professor, The University of Texas at Austin

Other Industry Experience

2024–2025 Visiting Researcher, Google (Research)
2022–2023 Visiting Researcher, Google (DeepMind)
2020–2021 Visiting Researcher, Google (Tensor Flow Ecosystem)

2019 Visiting Researcher, Facebook (AR Silicon Team)

2017–2018 Visiting Researcher, Google (gChips)

2015–2016 Consultant, Intel

2015–2016 Consultant, Advanced Micro Devices (AMD)

2014 Consultant, Intel

2010–2011 Senior Design Engineer, Advanced Micro Devices (AMD)

2009 Research Intern, Microsoft Research

2007–2009 Research Intern, VMware

2003–2006 Research Intern, Intel

Broader Impacts

Machine Learning Systems (MLSysBook.AI)

Open-access textbook and curriculum ecosystem for AI engineering education.

Reach 50+ universities across 5 continents teach from the curriculum; 22,800+ GitHub stars; 95+ contributors. Goal: 1M learners by 2030.

Scope 32 chapters (Vol. I: Foundations, Vol. II: At Scale), 35+ lecture decks, 20 TinyTorch modules, and 9,000+ interview questions in the StaffML bank.

Books Two MIT Press hardcover volumes forthcoming (2026, 2027).

Global TinyML4D Academic Network with ICTP: curriculum, mentorship, and hardware kits delivered to universities across Latin America, Africa, and Asia. Annual SciTinyML workshop series.

MLPerf & MLCommons (mlcommons.org)

Co-founded and serve as Vice President of MLCommons, the industry-standard ML benchmarking consortium adopted by Google, Microsoft, NVIDIA, Meta, AMD, Intel, and others.

2019/– Vice President and Board of Directors, MLCommons.

2019/– Research Co-Chair, MLCommons.

Impact Co-led the MLPerf Inference benchmark, now the industry standard across datacenter, edge, mobile, and IoT. MLPerf Inference selected for inclusion in the ISCA@50 25-Year Retrospective.

TinyML on edX

Founded and led the TinyML Professional Certificate on HarvardX/edX in partnership with Google and the tinyML Foundation, with **100,000+ learners** reached worldwide. Recognized as one of ClassCentral's 100 Most Popular Free Online Courses (2021) and winner of the CogX Best Course in AI award (2021).

Advisory & Board Roles

2024/– Board Member, EDGE AI Foundation (edgeaifoundation.org)

2020–2023 Member, International Roadmap for Devices and Systems (IRDS)

2019/– Vice President & Board of Directors, MLCommons (mlcommons.org)

2019/– Research Co-Chair, MLCommons

2019–2020 MLPerf Tiny Co-Chair, MLPerf (mlperf.org)

2018–2020 MLPerf Inference Co-Chair, MLPerf

2017–2020 Associate Editor, SIGARCH Blog (sigarch.org/blog)

Honors and Awards

2026 IEEE Micro Top Picks in Computer Architecture

2025 Best of Computer Architecture Letters (CAL), Editorial Board of IEEE CAL

2025 IEEE Micro Top Picks in Computer Architecture

2024 Best Paper Award, Vail Computer Elements Workshop (VCEW)

2023 MLPerf Inference selected for inclusion in ISCA@50 25-Year Retrospective

2023 IEEE Micro Top Picks in Computer Architecture (Honorable Mention)

2022 IEEE Micro Top Picks in Computer Architecture (Honorable Mention)

2021 BenchCouncil Rising Star Award, International Open Benchmark Council

2021 Deploying TinyML on HarvardX/edX: 100 Most Popular Free Online Courses, ClassCentral

2021 Best Course in AI: Tiny Machine Learning (TinyML) on HarvardX/edX, CogX Awards

2021 IEEE Micro Top Picks in Computer Architecture

2021 Best of Computer Architecture Letters (CAL), Editorial Board of IEEE CAL

2020 Programming Languages Software Award, ACM SIGPLAN

2020 Best Research Paper Award, Design Automation Conference (DAC)

2020 Google Faculty Research Award, Google

2019 Best Paper Nominee, IEEE International Symposium on Perf. Analysis of Systems and Software (ISPASS)

2019 Intl Symp. on High-Performance Computer Architecture (HPCA) Hall of Fame

2018 International Symp. on Microarchitecture (MICRO) Hall of Fame

2018 ACM SIGARCH CS TCCA Outstanding Dissertation Award (Advisee: Yuhao Zhu)

2017 IEEE Micro Top Picks in Computer Architecture

2017 Best Paper Nominee, Design Automation Conference (DAC)

2017 Google Faculty Research Award

2016 IEEE TCCA Young Computer Architect Award

2016 IEEE Micro Top Picks in Computer Architecture (Honorable Mention)

2016 Gilbreth Lectureship Honor, National Academy of Engineering (NAE)

2015 ACM SIGPLAN Most Influential PLDI Paper Award

2015 Google Faculty Research Award

2014 Best of Computer Architecture Letters (CAL) Award

2014 Best Paper Nominee, IEEE International Symposium on Microarchitecture Low Power Electronics and Design (ISLPED)

2014 Indo-American Frontiers of Engineering, National Academy of Engineering (NAE)

2013 Google Faculty Research Award

2013 Intel Early Career Award

2012 Google Faculty Research Award
2011 IEEE Micro Top Picks in Computer Architecture
2010 IEEE Micro Top Picks in Computer Architecture
2009 Best Paper Award, International Symposium on High-Performance Computer Architecture (HPCA)
2008 John A. and Elizabeth S. Armstrong Fellowship, Harvard University
2007 Best Student Presentation, International Symposium on Code Generation and Optimization (CGO)
2006 IEEE Micro Top Picks in Computer Architecture
2005 Best Paper Award, International Symposium on Microarchitecture (MICRO)
2003 Faculty Recognition for Technical Excellence, Santa Clara University
2003 Outstanding Undergraduate (Honorable), Computing Research Association (CRA)

University Committee Assignments

Harvard John A. Paulson School of Engineering and Applied Sciences

2023–2024 Member, Engineering Sciences Committee on Higher Degrees
2023 Member, Generative AI Steering Committee
2020 Member, Quantum Faculty Recruiting Committee
2019 Member, Robotics Faculty Recruiting Committee
2019/– Graduate Student Admissions Committee

The University of Texas at Austin

2016 Member, Faculty Recruiting Committee
2015 Member, Faculty Recruiting Committee
2014 Member, Technology in Teaching
2013 Member, Faculty Recruiting Committee
2011–2016 Graduate Student Admissions Committee

Professional Society Memberships

Institute of Electrical and Electronics Engineers (IEEE)
Association for Computing Machinery (ACM)

Professional Service

General Chair

2023 Tiny Machine Learning Research Symposium (TinyML)
2022 Tiny Machine Learning Research Symposium (TinyML)
2021 Tiny Machine Learning Research Symposium (TinyML)
2017 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)

Program Chair

2019 IEEE International Symposium on Workload Characterization (IISWC)

2014 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)

Program Committee

2025 Conference on Neural Information Processing Systems (NeurIPS)
2025 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2024 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2023 IEEE International Symposium on High-Performance Computer Architecture (HPCA)
2023 International Symposium on Computer Architecture (ISCA)
2023 Sixth Conference on Machine Learning and Systems (MLSys)
2023 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2022 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)
2022 International Symposium on Computer Architecture (ISCA)
2022 Fifth Conference on Machine Learning and Systems (MLSys)
2021 ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)
2021 IEEE Micro Top Picks
2021 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)
2021 IEEE International Symposium on High-Performance Computer Architecture (HPCA)
2020 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)
2020 International Symposium on Computer Architecture (ISCA)
2020 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2019 ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) (ERC)
2019 IEEE Micro Top Picks
2019 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)
2019 International Symposium on Computer Architecture (ISCA)
2019 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2018 International Symposium on Computer Architecture (ISCA) (ERC)
2018 IEEE Micro Top Picks
2018 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2018 IEEE International Symposium on Workload Characterization (IISWC)
2017 IEEE International Symposium on High-Performance Computer Architecture (HPCA) (ERC)
2017 International Symposium on Computer Architecture (ISCA)
2017 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2016 IEEE International Symposium on High-Performance Computer Architecture (HPCA) (ERC)
2016 IEEE Micro Top Picks
2016 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)
2016 IEEE International Symposium on Workload Characterization (IISWC)
2015 International Symposium on Computer Architecture (ISCA) (ERC)

2015 IEEE International Symposium on High-Performance Computer Architecture (HPCA)
2015 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)
2015 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2015 International Symposium on Performance Analysis of Systems and Software (ISPASS)
2015 International Symposium on Principles and Practice of Parallel Computing (PPoPP)
2015 IEEE International Symposium on Workload Characterization (IISWC)
2014 IEEE International Symposium on High-Performance Computer Architecture (HPCA)
2014 IEEE/ACM International Symposium on Code Generation and Optimization (CGO)
2014 IEEE/ACM International Symposium on Microarchitecture (MICRO)
2013 IEEE International Symposium on Workload Characterization (IISWC)

Organizing Committee

2023 Data-centric Machine Learning Research (DMLR) Workshop (ICML)
2020 ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)
2017 Design Automation Conference
2016–2017 Workshop on Cognitive Edge Computing (CogEdge)
2016 Tutorial on Tools for Mobile Computer Architecture (MobiTools)
2015 IEEE/ACM International Symposium on Code Generation and Optimization
2016 IEEE International Symposium on Microarchitecture
2015–2016 Tutorial on Simulation and Analysis Engine (ISCA, ASPLOS, HPCA, ICS, IISWC, ISPASS)
2013 IEEE International Symposium on Workload Characterization
2012 IEEE International Symposium on Performance Analysis of Systems and Software
2013 IEEE International Symposium on Performance Analysis of Systems and Software.

Guest Editor

2023 IEEE Micro Special Issue on Tiny Machine Learning
2016 IEEE Micro Special Issue on Internet of Things
2013 IEEE Micro Special Issue on Reliability-Aware Microarchitecture Design

Local Arrangements Chair

2013 Intl. Symp. on Performance Analysis of Systems and Software (ISPASS)
2015, 2016 Workshop on Silicon Errors in Logic: System Effects (SELSE).

Publications Chair

2013 Intl. Symp. on Workload Characterization (IISWC)

Community Activities

Tiny Machine Learning Open Education Initiative, https://tinyml.seas.harvard.edu (Founder)
Hands-on Computer Science (HaCS) for Austin Independent School District (via UT Outreach),

https://outreach.uteach.utexas.edu/uteach-outreach-cs-service-learning-program

Publications

Conference Publications

[FAccT'26] Jessica Quaye, Alicia Parrish, Charvi Rastogi, Minsuk Kahng, Oana Inel, Lora Aroyo, and Vijay Janapa Reddi. "Bridging the Scale Gap: Augmenting Human Red-Teaming to Uncover Latent Risks in T2I Models." In *FAccT 2026*. [DOI]

[FAccT'26] Charvi Rastogi, Mukul Bhutani, Minsuk Kahng, Shamsuddeen Hassan Muhammad, Evgeniia Razumovskaia, Priyanka Suresh, Ibrahim Said Ahmad, Charu Kalia, Yaaseen Mahomed, Madhurima Maji, Minjae Lee, Alicia Parrish, Jessica Quaye, Vijay Janapa Reddi, Aishwarya Verma, and Lora Aroyo. "Going PLACES: Participatory Localized Red Teaming for Text-to-Image Safety in the Global South." In *FAccT 2026*. [DOI]

[ASPLOS'26] Shvetank Prakash, Andrew Cheng, Olof Kindgren, Ashiq Ahamed, Graham Knight, Jedrzej Kufel, Francisco Rodriguez, Arya Tschand, David Kong, Mariam Elgamal, Jerry Huang, Emma Chen, Gage Hills, Richard Price, Emre Ozer, and Vijay Janapa Reddi. "Lifetime-Aware Design for Item-Level Intelligence at the Extreme Edge." In *ASPLOS 2026*. [DOI]

[DAC'25] David Kong, Shvetank Prakash, Jedrzej Kufel, Georgios Kyriazidis, Yasmine Omri, David Verity, Emre Ozer, Vijay Janapa Reddi, and Gage Hills. "333-eDRAM - 3T Embedded DRAM Leveraging Monolithic 3D Integration of 3 Transistor Types: IGZO, Carbon Nanotube and Silicon FETs." In *DAC 2025*. [DOI]

[HPCA'25] Subhankar Pal, Aporva Amarnath, Behzad Boroujerdian, Augusto Vega, Alper Buyuktosunoglu, John-David Wellman, Vijay Janapa Reddi, and Pradip Bose. "ARTEMIS: Agile Discovery of Efficient Real-Time Systems-on-Chips in the Heterogeneous Era." In *HPCA 2025*. [DOI]

[NeurIPS'25] Korneel Van den Berghe, Stein Stroobants, Vijay Janapa Reddi, and Guido de Croon. "Adaptive Surrogate Gradients for Sequential Reinforcement Learning in Spiking Neural Networks." In *NeurIPS 2025*. [Link]

[ISPASS'25] Zishen Wan, Jiayi Qian, Yuhang Du, Jason Jabbour, Yilun Du, Yang Zhao, Arijit Raychowdhury, Tushar Krishna, and Vijay Janapa Reddi. "Generative AI in Embodied Systems: System-Level Analysis of Performance, Efficiency and Scalability." In *ISPASS 2025*. [DOI]

[HPCA'25] Arya Tschand, Arun Tejusve Raghunath Rajan, Sachin Idgunji, Anirban Ghosh, Jeremy Holleman, Csaba Király, Pawan Ambalkar, Ritika Borkar, Ramesh Chukka, Trevor Cockrell, Oliver Curtis, Grigori Fursin, Miro Hodak, Hiwot Kassa, Anton Lokhmotov, Dejan Miskovic, Yuechao Pan, Manu Prasad Manmathan, Liz Raymond, Tom St. John, Arjun Suresh, Rowan Taubitz, Sean Zhan, Scott Wasson, David Kanter, and Vijay Janapa Reddi. "MLPerf Power: Benchmarking the Energy Efficiency of Machine Learning Systems from \(μ\)Watts to MWatts for Sustainable AI." In *HPCA 2025*. [DOI]

[ASPLOS'25] Peiqing Chen, Minghao Li, Zishen Wan, Yu-Shun Hsiao, Minlan Yu, Vijay Janapa Reddi, and Zaoxing Liu. "OctoCache: Caching Voxels for Accelerating 3D Occupancy Mapping in Autonomous Systems." In *ASPLOS 2025*. [DOI]

[ASPLOS'25] Zishen Wan, Yuhang Du, Mohamed Ibrahim, Jiayi Qian, Jason Jabbour, Yang Katie Zhao, Tushar

Krishna, Arijit Raychowdhury, and Vijay Janapa Reddi. "ReCA: Integrated Acceleration for Real-Time and Efficient Cooperative Embodied Autonomous Agents." In *ASPLOS 2025*. [DOI]

[FAccT'24] Jessica Quaye, Alicia Parrish, Oana Inel, Charvi Rastogi, Hannah Rose Kirk, Minsuk Kahng, Erin van Liemt, Max Bartolo, Jess Tsang, Justin White, Nathan Clement, Rafael Mosquera, Juan Ciro, Vijay Janapa Reddi, and Lora Aroyo. "Adversarial Nibbler: An Open Red-Teaming Method for Identifying Diverse Harms in Text-to-Image Generation." In *FAccT 2024*. [DOI]

[ASPLOS'24] Maximilian Lam, Jeff Johnson, Wenjie Xiong, Kiwan Maeng, Udit Gupta, Yang Li, Liangzhen Lai, Ilias Leontiadis, Minsoo Rhu, Hsien-Hsin S. Lee, Vijay Janapa Reddi, Gu-Yeon Wei, David Brooks, and G. Edward Suh. "GPU-based Private Information Retrieval for On-Device Machine Learning Inference." In *ASPLOS 2024*. [DOI]

[ICCAD'24] Jason Jabbour and Vijay Janapa Reddi. "Generative AI Agents in Autonomous Machines: A Safety Perspective." In *ICCAD 2024*. [DOI]

[BioCAS'24] Biyan Zhou, Pao-Sheng Vincent Sun, Jason Yik, Charlotte Frenkel, Vijay Janapa Reddi, and Arindam Basu. "Grand Challenge on Neural Decoding for Motor Control of non-Human Primates." In *BioCAS 2024*. [DOI]

[DAC'24] Sabrina M. Neuman, Brian Plancher, and Vijay Janapa Reddi. "Invited: The Magnificent Seven Challenges and Opportunities in Domain-Specific Accelerator Design for Autonomous Systems." In *DAC 2024*. [DOI]

[ISSS'24] Vijay Janapa Reddi. "MLSysBook.AI: Principles and Practices of Machine Learning Systems Engineering." In *ISSS 2024*. [DOI]

[ASPLOS'24] Zishen Wan, Nandhini Chandramoorthy, Karthik Swaminathan, Pin-Yu Chen, Kshitij Bhardwaj, Vijay Janapa Reddi, and Arijit Raychowdhury. "MulBERRY: Enabling Bit-Error Robustness for Energy-Efficient Multi-Agent Autonomous Systems." In *ASPLOS 2024*. [DOI]

[ICRA'24] Víctor Mayoral Vilches, Jason Jabbour, Yu-Shun Hsiao, Zishen Wan, Martiño Crespo-Álvarez, Matthew Stewart, Juan Manuel Reina-Muñoz, Prateek Nagras, Gaurav Vikhe, Mohammad Bakhshalipour, Martin Pinzger, Stefan Rass, Smruti Panigrahi, Giulio Corradi, Niladri Roy, Phillip B. Gibbons, Sabrina M. Neuman, Brian Plancher, and Vijay Janapa Reddi. "RobotPerf: An Open-Source, Vendor-Agnostic, Benchmarking Suite for Evaluating Robotics Computing System Performance." In *ICRA 2024*. [DOI]

[AAAI'24] Brian Plancher, Sebastian Büttrich, Jeremy Ellis, Neena Goveas, Laila D. Kazimierski, Jesús Alfonso López Sotelo, Milan Lukic, Diego Mendez, Rosdiadee Nordin, Andrés Oliva Trevisan, Massimo Pavan, Manuel Roveri, Marcus Rüb, Jackline Tum, Marian Verhelst, Salah Abdeljabar, Segun Adebayo, Thomas Amberg, Halleluyah Oluwatobi Aworinde, José Bagur, Gregg Barrett, Nabil Benamar, Bharat S. Chaudhari, Ronald Criollo, David Cuartielles, José A. Ferreira Filho, Solomon Gizaw, Evgeni Gousev, Alessandro Grande, Shawn Hymel, Peter Ing, Prashant Manandhar, Pietro Manzoni, Boris Murmann, Eric Pan, Rytis Paskauskas, Ermanno Pietrosemoli, Tales C. Pimenta, Marcelo Rovai, Marco Zennaro, and Vijay Janapa Reddi. "TinyML4D: Scaling Embedded Machine Learning Education in the Developing World." In *AAAI 2024*. [DOI]

[ISCA'23] Srivatsan Krishnan, Amir Yazdanbakhsh, Shvetank Prakash, Jason Jabbour, Ikechukwu Uchendu, Susobhan Ghosh, Behzad Boroujerdian, Daniel Richins, Devashree Tripathy, Aleksandra Faust, and Vijay Janapa Reddi. "ArchGym: An Open-Source Gymnasium for Machine Learning Assisted Architecture Design." In *ISCA 2023*. [DOI]

[DAC'23] Vijay Janapa Reddi and Amir Yazdanbakhsh. "Architecture 2.0: Challenges and Opportunities." In *DAC 2023*. [DOI]

[DAC'23] Zishen Wan, Nandhini Chandramoorthy, Karthik Swaminathan, Pin-Yu Chen, Vijay Janapa Reddi, and Arijit Raychowdhury. "BERRY: Bit Error Robustness for Energy-Efficient Reinforcement Learning-Based Autonomous Systems." In *DAC 2023*. [DOI]

[ISPASS'23] Shvetank Prakash, Tim Callahan, Joseph Bushagour, Colby R. Banbury, Alan V. Green, Pete Warden, Tim Ansell, and Vijay Janapa Reddi. "CFU Playground: Full-Stack Open-Source Framework for Tiny Machine Learning (TinyML) Acceleration on FPGAs." In *ISPASS 2023*. [DOI]

[DATE'23] Shvetank Prakash, Tim Callahan, Joseph Bushagour, Colby R. Banbury, Alan V. Green, Pete Warden, Tim Ansell, and Vijay Janapa Reddi. "CFU Playground: Want a faster ML processor? Do it yourself!." In *DATE 2023*. [DOI]

[NeurIPS'23] Mark Mazumder, Colby R. Banbury, Xiaozhe Yao, Bojan Karlas, William Gaviria Rojas, Sudnya Frederick Diamos, Greg Diamos, Lynn He, Alicia Parrish, Hannah Rose Kirk, Jessica Quaye, Charvi Rastogi, Douwe Kiela, David Jurado, David Kanter, Rafael Mosquera, Will Cukierski, Juan Ciro, Lora Aroyo, Bilge Acun, Lingjiao Chen, Mehul Raje, Max Bartolo, Evan Sabri Eyuboglu, Amirata Ghorbani, Emmett D. Goodman, Addison Howard, Oana Inel, Tariq Kane, Christine R. Kirkpatrick, D. Sculley, Tzu-Sheng Kuo, Jonas W. Mueller, Tristan Thrush, Joaquin Vanschoren, Margaret Warren, Adina Williams, Serena Yeung, Newsha Ardalani, Praveen K. Paritosh, Ce Zhang, James Y. Zou, Carole-Jean Wu, Cody Coleman, Andrew Y. Ng, Peter Mattson, and Vijay Janapa Reddi. "DataPerf: Benchmarks for Data-Centric AI Development." In *NeurIPS 2023*. [Link]

[MLSys'23] Colby R. Banbury, Vijay Janapa Reddi, Alexander Elium, Shawn Hymel, David Tischler, Daniel Situnayake, Carl Ward, Louis Moreau, Jenny Plunkett, Matthew Kelcey, Mathijs Baaijens, Alessandro Grande, Dmitry Maslov, Arthur Beavis, Jan Jongboom, and Jessica Quaye. "Edge Impulse: An MLOps Platform for Tiny Machine Learning." In *MLSys 2023*. [Link]

[DATE'23] Yu-Shun Hsiao, Zishen Wan, Tianyu Jia, Radhika Ghosal, Abdulrahman Mahmoud, Arijit Raychowdhury, David Brooks, Gu-Yeon Wei, and Vijay Janapa Reddi. "MAVFI: An End-to-End Fault Analysis Framework with Anomaly Detection and Recovery for Micro Aerial Vehicles." In *DATE 2023*. [DOI]

[ISCA'23] Sabrina M. Neuman, Radhika Ghosal, Thomas Bourgeat, Brian Plancher, and Vijay Janapa Reddi. "RoboShape: Using Topology Patterns to Scalably and Flexibly Deploy Accelerators Across Robots." In *ISCA 2023*. [DOI]

[ICCS'23] Thanh Thi Nguyen, Minh Cuong Nguyen, Thien Huynh-The, Quoc-Viet Pham, Quoc Viet Hung Nguyen, Imran Razzak, and Vijay Janapa Reddi. "Solving Complex Sequential Decision-Making Problems by Deep Reinforcement Learning with Heuristic Rules." In *ICCS 2023*. [DOI]

[IROS'23] Yu-Shun Hsiao, Siva Kumar Sastry Hari, Balakumar Sundaralingam, Jason Yik, Thierry Tambe, Charbel Sakr, Stephen W. Keckler, and Vijay Janapa Reddi. "VaPr: Variable-Precision Tensors to Accelerate Robot Motion Planning." In *IROS 2023*. [DOI]

[MLSys'23] Hyoukjun Kwon, Krishnakumar Nair, Jamin Seo, Jason Yik, Debabrata Mohapatra, Dongyuan Zhan, Jinook Song, Peter Capak, Peizhao Zhang, Peter Vajda, Colby R. Banbury, Mark Mazumder, Liangzhen Lai, Ashish Sirasao, Tushar Krishna, Harshit Khaitan, Vikas Chandra, and Vijay Janapa Reddi. "XRBench: An Extended Reality (XR) Machine Learning Benchmark Suite for the Metaverse." In *MLSys 2023*. [Link]

[MICRO'22] Srivatsan Krishnan, Zishen Wan, Kshitij Bhardwaj, Paul N. Whatmough, Aleksandra Faust, Sabrina M. Neuman, Gu-Yeon Wei, David Brooks, and Vijay Janapa Reddi. "Automatic Domain-Specific SoC Design for Autonomous Unmanned Aerial Vehicles." In *MICRO 2022*. [DOI]

[DATE'22] Zishen Wan, Aqeel Anwar, Abdulrahman Mahmoud, Tianyu Jia, Yu-Shun Hsiao, Vijay Janapa Reddi, and Arijit Raychowdhury. "FRL-FI: Transient Fault Analysis for Federated Reinforcement Learning-Based Navigation Systems." In *DATE 2022*. [DOI]

[ICRA'22] Brian Plancher, Sabrina M. Neuman, Radhika Ghosal, Scott Kuindersma, and Vijay Janapa Reddi. "GRiD: GPU-Accelerated Rigid Body Dynamics with Analytical Gradients." In *ICRA 2022*. [DOI]

[MLSys'22] Vijay Janapa Reddi, David Kanter, Peter Mattson, Jared Duke, Thai Nguyen, Ramesh Chukka, Kenneth Shiring, Koan-Sin Tan, Mark Charlebois, William Chou, Mostafa El-Khamy, Jungwook Hong, Tom St. John, Cindy Trinh, Michael Buch, Mark Mazumder, Relja Markovic, Thomas Atta-Fosu, Fatih Çakir, Masoud Charkhabi, Xiaodong Chen, Cheng-Ming Chiang, Dave Dexter, Terry Heo, Guenther Schmuelling, Maryam Shabani, and Dylan Zika. "MLPerf Mobile Inference Benchmark: An Industry-Standard Open-Source Machine Learning Benchmark for On-Device AI." In *MLSys 2022*. [Link]

[DATE'22] Tianyu Jia, En-Yu Yang, Yu-Shun Hsiao, Jonathan J. Cruz, David Brooks, Gu-Yeon Wei, and Vijay Janapa Reddi. "OMU: A Probabilistic 3D Occupancy Mapping Accelerator for Real-time OctoMap at the Edge." In *DATE 2022*. [DOI]

[IROS'22] Víctor Mayoral Vilches, Sabrina M. Neuman, Brian Plancher, and Vijay Janapa Reddi. "RobotCore: An Open Architecture for Hardware Acceleration in ROS 2." In *IROS 2022*. [DOI]

[AICAS'22] Zishen Wan, Ashwin Sanjay Lele, Bo Yu, Shaoshan Liu, Yu Wang, Vijay Janapa Reddi, Cong Hao, and Arijit Raychowdhury. "Robotic Computing on FPGAs: Current Progress, Research Challenges, and Opportunities." In *AICAS 2022*. [DOI]

[ISPASS'22] Srivatsan Krishnan, Zishen Wan, Kshitij Bhardwaj, Ninad Jadhav, Aleksandra Faust, and Vijay Janapa Reddi. "Roofline Model for UAVs: A Bottleneck Analysis Tool for Onboard Compute Characterization of Autonomous Unmanned Aerial Vehicles." In *ISPASS 2022*. [DOI]

[NeurIPS'22] William A. Gaviria Rojas, Sudnya Frederick Diamos, Keertan Kini, David Kanter, Vijay Janapa Reddi, and Cody Coleman. "The Dollar Street Dataset: Images Representing the Geographic and Socioeconomic Diversity of the World." In *NeurIPS 2022*. [Link]

[AICAS'22] Sabrina M. Neuman, Brian Plancher, Bardienus Pieter Duisterhof, Srivatsan Krishnan, Colby R. Banbury, Mark Mazumder, Shvetank Prakash, Jason Jabbour, Aleksandra Faust, Guido C. H. E. de Croon, and Vijay Janapa Reddi. "Tiny Robot Learning: Challenges and Directions for Machine Learning in Resource-Constrained Robots." In *AICAS 2022*. [DOI]

[SIGCSE'22] Brian Plancher and Vijay Janapa Reddi. "TinyMLedu: The Tiny Machine Learning Open Education Initiative." In *SIGCSE 2022*. [DOI]

[DAC'22] Yu-Shun Hsiao, Siva Kumar Sastry Hari, Michal Filipiuk, Timothy Tsai, Michael B. Sullivan, Vijay Janapa Reddi, Vasu Singh, and Stephen W. Keckler. "Zhuyi: perception processing rate estimation for safety in autonomous vehicles." In *DAC 2022*. [DOI]

[ISPASS'21] Michael Buch, Zahra Azad, Ajay Joshi, and Vijay Janapa Reddi. "AI Tax in Mobile SoCs: End-to-end Performance Analysis of Machine Learning in Smartphones." In *ISPASS 2021*. [DOI]

[DAC'21] Zishen Wan, Aqeel Anwar, Yu-Shun Hsiao, Tianyu Jia, Vijay Janapa Reddi, and Arijit Raychowdhury. "Analyzing and Improving Fault Tolerance of Learning-Based Navigation Systems." In *DAC 2021*. [DOI]

[Interspeech'21] Mark Mazumder, Colby R. Banbury, Josh Meyer, Pete Warden, and Vijay Janapa Reddi. "Few-Shot Keyword Spotting in Any Language." In *22nd Annual Conference of the International Speech Communica 2021*. [DOI]

[ICML'21] Maximilian Lam, Gu-Yeon Wei, David Brooks, Vijay Janapa Reddi, and Michael Mitzenmacher. "Gradient Disaggregation: Breaking Privacy in Federated Learning by Reconstructing the User Participant Matrix." In *ICML 2021*. [Link]

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[arXiv'21] Mark Mazumder, Colby R. Banbury, Josh Meyer, Pete Warden, and Vijay Janapa Reddi. "Few-Shot Keyword Spotting in Any Language." In *arXiv:2104.01454*. [Link]

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Workshop Papers

[HotStorage’17] Jayashree Mohan, Dhathri Purohith, Matthew Halpern, Vijay Chidambaram, and Vijay Janapa Reddi. “Storage on Your SmartPhone Uses More Energy Than You Think.” In *HotStorage 2017*. [Link]

[INTERACT’05] Alex Shye, Matthew Iyer, Tipp Moseley, David Hodgdon, Dan Fay, Vijay Janapa Reddi, and Daniel A. Connors. “Analysis of path profiling information generated with performance monitoring hardware.” In *INTERACT-9 2005*. [DOI]

[AADEBUG’05] Alex Shye, Matthew Iyer, Vijay Janapa Reddi, and Daniel A. Connors. “Code coverage testing using hardware performance monitoring support.” In *AADEBUG 2005*. [DOI]

[ISCA’04] Vijay Janapa Reddi, Alex Settle, Daniel A. Connors, and Robert S. Cohn. “PIN: a binary instrumentation tool for computer architecture research and education.” In *ISCA 2004*. [DOI]

Books and Chapters

[Book’13] Vijay Janapa Reddi and Meeta Sharma Gupta. “Resilient Architecture Design for Voltage Variation.” In *Synthesis Lectures on Computer Architecture, 2013*. [DOI]

Talks

10/2024 “MLSysBook.AI: Principles and Practices of Machine Learning Systems Engineering.” Embedded Systems Week (ESWEEK), Virtual.

09/2024 “Architecture 2.0.” International Symposium on Workload Characterization (IISWC). (Keynote).

9/2024 “Architecture 2.0: Challenges and Opportunities for tinyML.” tinyAI Forum on Generative AI on the Edge, Virtual.

6/2024 “Architecture 2.0.” Design Automation Conference (DAC).

6/2024 “The Magnificent Seven Challenges and Opportunities in Domain-Specific Accelerator Design for Autonomous Systems.” Design Automation Conference (DAC).

6/2024 “Architecture 2.0: From Concept to Breaking Ground.” CogArch Workshop at (ISCA). (Keynote).

6/2024 “The Magnificent Seven Challenges and Opportunities in Domain-Specific Accelerator Design for Autonomous Systems.” EPOCHS Workshop at ISCA.

4/2024 “The State of TinyML Benchmarking: Current Landscape, Challenges, and Emerging Trends.” tinyML Summit.

1/2024 "Architecture 2.0." Free and Open Source Silicon (FOSSI).

10/2023 "Architecture 2.0." Open Compute Project (OCP), Virtual.

09/2023 "Architecture 2.0." MIT Industry-Academia Partnership.

07/2023 "Architecture 2.0." Design Automation Conference (DAC), Virtual.

06/2023 "The Parameter and Chip Wars." Vail Computer Elements Workshop, Virtual.

06/2023 "Adopting AI: With Power Comes Responsibility." Panel, FDCAI, Virtual.

04/2023 "TinyML." IEEE International Symposium on Low-Power and High-Speed Chips (COOL Chips), Virtual.

04/2023 "The Parameter and Chip Wars: Shifting the Focus from Model-centric to Data-centric AI." MICRON, Virtual.

04/2023 "NeuroBench: Advancing Neuromorphic Computing through Collaborative and Rigorous Benchmarking." NICE, Virtual.

11/2022 "TinyML." Urban Sensor Networks Workshop Panel, Virtual.

10/2022 "ML Metrics: The Past, Present, and Future of Benchmarking ML Systems, Datasets, and Use Cases." Specialization with Benchmarks for Emerging Applications (MICRO), Virtual.

09/2022 "Tiny Machine Learning." Chips & Compilers Symposium, MLSys '22, Virtual.

09/2022 "MLPerf & DataPerf." The Autonomous Vehicle Computing Consortium (AVCC), Virtual.

09/2022 "Benchmarking FastML Systems." Fast ML for Science Workshop.

08/2022 "Tiny Machine Learning: A System-level Perspective." ACM/IEEE International Symposium on Low Power Electronics and Design, Virtual. (Keynote).

08/2022 "The Vision Behind MLPerf and DataPerf." Monterey Data Conference, Virtual.

08/2022 "DataPerf: Benchmarks for Data-centric AI Development." The Future of Data-Centric AI, Snorkel.ai, Virtual.

07/2022 "Tiny Machine Learning: Challenges and Opportunities." Design Automation Conference, ROAD4NN Workshop, Virtual. (Keynote).

07/2022 "The Future of Smart Cities is Tiny and Bright." ACM International Conference on Future Energy Systems (ACM e-Energy), Virtual. (Keynote).

06/2022 "Machine Learning Metrics." HiPEAC AccML Workshop.

05/2022 "Democratizing TinyML." Rutgers Efficient AI (REFAI) Seminar, Virtual.

04/2022 "Tiny Machine Learning (TinyML) for Domain-Specific Systems." International Workshop on Domain Specific System Architecture (DOSSA-4), Virtual.

03/2022 "IoT 2.0: The Era of Intelligence on Things." Design Automation and Test in Europe, Virtual.

01/2022 "Tiny Machine Learning." Accelerated Machine Learning Workshop, co-located with HiPEAC 2022, Virtual.

12/2021 "Machine Learning's Future is Tiny & Bright." AICON GWANGJU, Virtual. (Keynote).

12/2021 "Democratizing TinyML." Globecom Workshop on Sustainable Environmental Sensing Systems, Virtual. (Keynote).

11/2021 "Tiny Machine Learning." EdukCircle International Convention on Engineering and Computer Technology, Virtual.

11/2021 "Tiny Machine Learning (TinyML) for Robotics." Conference on Robot Learning, Virtual. (Keynote).

11/2021 "Democratizing TinyML: Generalization, Standardization and Automation." Workshop on Hardware and Algorithms for Learning On-a-chip (HALO) workshop, ICCAD conference, Virtual. (Keynote).

11/2021 "Democratizing TinyML: Generalization, Standardization and Automation." Multi-DNN Workshop, co-located with MICRO, Virtual. (Keynote).

11/2021 "Data for TinyML." Data for AI Summit @Google (internal), Virtual. (Keynote).

10/2021 "Widening Access to Applied Machine Learning with TinyML." IEEE Global Humanitarian Technology Conference, Virtual. (Keynote).

10/2021 "The Vision Behind MLPerf." SiFive Engineering Forum, Virtual.

10/2021 "The Vision Behind MLPerf." Samsung AI Cambridge, Santa Clara.

10/2021 "Democratizing TinyML." MICRO 2021 Workshop on Systems for Multi-DNN Workloads, Virtual. (Keynote).

09/2021 "The Vision Behind MLPerf." Tensorrent, Virtual.

07/2021 "Tiny Machine Learning." Workshop on Artificial Intelligence, Machine Learning, & Computational Intelligence, Virtual.

07/2021 "The Vision Behind MLPerf." STMicroelectronics, Virtual.

03/2021 "tinyMLPerf: Benchmarking Ultra-low-power Systems." Tiny Machine Learning Summit, San Francisco.

03/2021 "tinyMLPerf: Benchmarking Ultra-low-power Systems." "Machine Learning at the Edge," Workshop co-located with Design Automation Conference.

06/2020 "The Vision Behind MLPerf." AMD Tech Talk, Austin.

03/2020 "tinyMLPerf: Benchmarking Ultra-low-power Systems." Tiny Machine Learning Summit.

02/2020 "The Vision Behind MLPerf." International Solid-State Circuits Conference (ISSCC), San Francisco.

02/2020 "MLPerf Inference." Machine Learning Systems Workshop, Santa Clara.

09/2019 "The Vision Behind MLPerf: A Broad ML Benchmark Suite for Measuring the Performance of ML Software Frameworks, ML Hardware Accelerators in Cloud and Edge Computing." Taiwan Semiconductor Manufacturing Company (TSMC).

09/2019 "The Vision Behind MLPerf: A Broad ML Benchmark Suite for Measuring the Performance of ML Software Frameworks, ML Hardware Accelerators in Cloud and Edge Computing." Taiwan AI Labs. (Keynote).

09/2019 "The Vision Behind MLPerf: A Broad ML Benchmark Suite for Measuring the Performance of ML Software Frameworks, ML Hardware Accelerators in Cloud and Edge Computing." Synopsis SNUG, Taiwan.

04/2019 "Ten Commandments for Mobile Computer Architecture." Workshop on Infrastructure and Methodology for SoC-level Performance and Power Modeling, co-located with ASPLOS.

03/2019 "The Vision Behind MLPerf (mlperf.org)." Intel VSSAD.

03/2019 "Evaluating Resiliency in End-to-end Learning for Autonomous Machines." The 15th Workshop on Silicon Errors in Logic – System Effects.

03/2019 "Autonomous Aerial Computing Machines." International Workshop on Performance Analysis of Machine Learning Systems.

12/2018 "The Vision Behind MLPerf: A Broad ML Benchmark Suite for Measuring the Performance of ML Software Frameworks, ML Hardware Accelerators, and ML Cloud and Edge Platforms." IEEE

BigBench co-located with the IEEE Big Data Conference. (Keynote).

12/2018 "The Vision Behind MLPerf: A Broad ML Benchmark Suite for Measuring the Performance of ML Software Frameworks, ML Hardware Accelerators in Cloud and Edge Computing." The Forum of Turing Centers, Shanghai Jiao Tong University. (Keynote).

12/2018 "The Vision Behind MLPerf: A Broad ML Benchmark Suite for Measuring the Performance of ML Software Frameworks, ML Hardware Accelerators in Cloud and Edge Computing." Boston Area Computer Architecture Workshop. (Keynote).

11/2018 "The Vision Behind MLPerf: A Broad ML Benchmark Suite for Measuring the Performance of ML Software Frameworks, ML Hardware Accelerators, and ML Cloud and Edge Platforms." Samsung Advanced Computing Lab (ACL).

10/2018 "The Vision Behind MLPerf: A Broad ML Benchmark Suite for Measuring the Performance of ML Software Frameworks, ML Hardware Accelerators, and ML Cloud and Edge Platforms." Samsung Austin Research Center (SARC).

10/2018 "Mobile Robotics for Computer Architects." First Annual Workshop on Domain Specific System Architecture co-located with International Symposium on Microarchitecture (MICRO). (Keynote).

04/2018 "Aerial Computing: Challenges and Opportunities for Hardware and Software Architects Designing Flying Systems." IBM T. J. Watson.

03/2018 "Architecting for Big Data Analytics: Think Dubai rather than Venice." Workshop on BigData Benchmarks, Performance, Optimization and Emerging Hardware (co-located with ASPLOS).

02/2017 "Architecture Support for Scripting from Mobile to Cloud." Stanford University, Palo Alto.

05/2016 "Watt-Wise-Web://Architecting for Responsiveness and Energy-Efficiency." The University of Chicago, Chicago–IL.

05/2016 "Mobile CPU Evolution: The Past, the Present, and the Future." Rice University – TexasWISE Keynote, Houston.

05/2016 "Microarchitectural Implications of Event-driven Programming." Northwestern, Chicago–IL.

05/2016 "Microarchitectural Implications of Event-driven Programming." Intel Santa Clara–CA.

05/2016 "Microarchitectural Implications of Event-driven Programming." AMD Austin–TX.

03/2016 "Programming the Web of Things: Why Architects Should Care." Sensors to Cloud Architectures Workshop, Barcelona. (Keynote).

02/2016 "From Moore's Law to Moore's Crawl: Architecting the Next-Generation of Mobile Computing Devices." University of Washington, Seattle–WA.

02/2016 "From Moore's Law to Moore's Crawl: Architecting the Next-Generation of Mobile Computing Devices." National Academy of Engineering (NAE) Annual Event, Irvine–CA.

12/2015 "Programming the Web of Things." Workshop on Internet of Things (IoT) held in conjunction with International Symposium on Microarchitecture, Hawaii.

12/2015 "End of the Road for My CAREER." Workshop on Negative Outcomes, Post-mortems, and Experiences (NOPE) held in conjunction with International Symposium on Microarchitecture, Hawaii.

11/2015 "Watt-Wise Web: Architecting for a Responsive and Energy-Efficient Mobile Web." Texas A&M University.

10/2015 "Watt-Wise Web: Architecting for a Responsive and Energy-Efficient Mobile Web." Google Faculty Summit.

10/2015 "Watt-Wise Web: Architecting for a Responsive and Energy-Efficient Mobile Web." Georgia Tech

University.

09/2015 "What Users Want and What Hardware Provides: Bridging the Gap Between User Quality of Experience (QoE) and Mobile Device Trends." National Taiwan University, Taiwan.

09/2015 "What Users Want and What Hardware Provides: Bridging the Gap Between User Quality of Experience (QoE) and Mobile Device Trends." Mediatek, Taiwan.

09/2015 "What Users Want and What Hardware Provides: Bridging the Gap Between User Quality of Experience (QoE) and Mobile Device Trends." Academia Sinica, Taiwan.

09/2015 "Mobile CPU Evolution: The Past, the Present, and the Future." Taiwan Application Processor Union – Mobile SoC Summer Course, Taiwan.

06/2015 "What Users Want and What Hardware Provides: Bridging the Gap Between User Quality of Experience (QoE) and Mobile Device Trends." Duke University, Raleigh–NC.

06/2015 "GPU Voltage Guardband Management to Achieve Exascale Energy-Efficiency." AMD Austin–TX.

05/2015 "Voltage Noise in Multicore Processors." Intel, Portland.

05/2015 "GPU Voltage Guardband Management to Achieve Exascale Energy-Efficiency." Intel, Portland.

04/2015 "What Users Want and What Hardware Provides: Bridging the Gap Between User Quality of Experience (QoE) and Mobile Device Trends." Qualcomm, Raleigh–NC.

04/2015 "Mobile CPU Evolution: The Past, the Present, and the Future." Microsoft, Seattle–WA.

03/2015 "What Users Want and What Hardware Provides: Bridging the Gap Between User Quality of Experience (QoE) and Mobile Device Trends." Facebook, Menlo Park–CA.

02/2015 "Mobile CPU Evolution: The Past, the Present, and the Future." Intel Santa Clara–CA.

11/2014 "Watt-Wise Web: Architecting for a Responsive and Energy-Efficient Mobile Web." Univ. of Michigan.

06/2014 "Simulators are Perfect, Authors are Oracles, Users are Innocent." Workshop on Duplicating, Deconstructing and Debunking (WDDD) held in conjunction with International Symposium on Computer Architecture.

06/2014 "Architecting for the Mobile Web: Where We've Been, Where We're Heading, and What We Need to Address." Parallelism in Mobile Platforms (PRISM) held in conjunction with International Symposium on Computer Architecture.

05/2014 "Mobile Processor Architectures: Design Implications and Challenges for Energy Efficiency." Indo-American Frontiers of Engineering (IAFOE), Mysore–India.

05/2014 "Hardware and Software Co-Design for Robust and Resilient Execution." International Conference on Integrated Circuit Design and Technology (ICICDT), Austin.

03/2014 "Architectural Support for the Interactive Mobile Web." Samsung Austin–TX.

03/2014 "Architectural Support for the Interactive Mobile Web." ARM Austin–TX.

02/2014 "Robust and Resilient Systems from the Bottom-Up: Circuits, Architecture and Software Integration." ISSCC Forum, San Francisco–CA.

02/2014 "Architectural Support for the Interactive Mobile Web." Intel Austin–TX.

02/2013 "Toward High-Performance and Energy-Efficient Mobile Web Browsing." Qualcomm, Santa Clara–MA.

08/2012 "Toward High-Performance and Energy-Efficient Mobile Web Browsing." Intel Austin–TX.

08/2012 "Toward High-Performance and Energy-Efficient Mobile Web Browsing." AMD Austin–TX.

10/2010 "Web Search Using Small Cores." AMD, Boxborough–MA.

07/2010 "Web Search Using Small Cores." SeaMicro Santa Clara–CA.

07/2010 "Web Search Using Small Cores." Intel Hudson–MA.

07/2010 "Web Search Using Small Cores." IBM T. J. Watson Labs, Hawthorne–NY.

07/2010 "Web Search Using Small Cores." HP Labs, Palo Alto–CA.

07/2010 "Web Search Using Small Cores." Google, Palo Alto–CA.

07/2010 "Web Search Using Small Cores." Facebook, Palo Alto–CA.

07/2010 "Software-Assisted Hardware Reliability." Intel, Portland.

07/2010 "Software-Assisted Hardware Reliability." IBM T. J. Watson Labs, Yorktown–NY.

06/2010 "Web Search Using Small Cores." Amazon, Seattle–WA.

06/2010 "Software-Assisted Hardware Reliability." Microsoft Research, Redmond–WA.

03/2010 "Software-Assisted Hardware Reliability." Intel Santa Clara–CA.

03/2010 "Software-Assisted Hardware Reliability." AMD Austin–TX.

03/2007 "Persistent Code Caching." Intel Santa Clara–CA.

Books

2027 V. Janapa Reddi. *Machine Learning Systems at Scale*. MIT Press. *Forthcoming.* (Volume II of the MLSysBook series.)

2026 V. Janapa Reddi. *Introduction to Machine Learning Systems*. MIT Press. *Forthcoming.* (Volume I of the MLSysBook series; companion site mlsysbook.ai.)

2013 V. Janapa Reddi and Meeta Sharma Gupta. *Resilient Architecture Design for Voltage Variation*. Synthesis Lectures on Computer Architecture. Morgan & Claypool Publishers. ISBN: 9781608456376.

Technical Reports

2010 V. Janapa Reddi, B. Lee, T. Chilimbi, K. Vaid. "Web Search Using Small Cores: Quantifying the Price of Efficiency," in Microsoft Research Tech. Report, June.

Theses

2010 V. Janapa Reddi. "Software-Assisted Hardware Reliability: Enabling Aggressive Timing Speculation Using Run-Time Feedback from Hardware and Software," Ph.D. Thesis, School of Engineering and Applied Sciences, Harvard University.

2005 V. Janapa Reddi. "Deploying Dynamic Code Transformation in Modern Computing Environments," M.S. Thesis, Department of Electrical and Computer Engineering, University of Colorado.

2003 V. Janapa Reddi. "Heterogeneous Networks of Workstations Across Wide Area Networks," B.S. Thesis, Department of Electrical and Computer Engineering, Santa Clara University.

Patents

2005 R. Cohn, T. Moseley, and V. Janapa Reddi. "System and method to instrument references to shared memory." U.S. Patent Application 11/143,130, filed June 1, 2005.

2012 N. Kim, J. O'Connor, M. Schulte, and V. Janapa Reddi. "Method and apparatus for power reduction during lane divergence." U.S. Patent Application 13/605,460, filed September 6, 2012.

2015 V. Janapa Reddi, M. Gupta, G. Holloway, G. Wei, M. D. Smith, and D. Brooks. "Adaptive event-guided system and method for avoiding voltage emergencies." U.S. Patent 8,949,666, issued February 3, 2015.

Teaching

Harvard University

Sp' 2026 ES 50: Introduction to Electrical and Computer Engineering

Sp' 2025 ES 50: Introduction to Electrical and Computer Engineering

Sp' 2024 COMPSCI 141: Computing Hardware

Fa' 2023 COMPSCI 249R: Tiny Machine Learning

Sp' 2023 COMPSCI 141: Computing Hardware

Fa' 2022 COMPSCI 249R: Tiny Machine Learning

Sp' 2021 COMPSCI 141: Computing Hardware

Fa' 2020 COMPSCI 249R: Tiny Machine Learning

Fa' 2019 COMPSCI 249R: Autonomous Machines

The University of Texas at Austin

Sp' 2016 EE 319K: Introduction to Embedded Systems

Sp' 2015 EE 319K: Introduction to Embedded Systems

Fa' 2014 EE 382V: Code Generation and Optimization

Sp' 2014 EE 319K: Introduction to Embedded Systems

Fa' 2013 EE 382V: Code Generation and Optimization

Sp' 2013 EE 319K: Introduction to Embedded Systems

Fa' 2012 EE 382V: Dynamic Compilation

Sp' 2012 EE 382V: Code Generation and Optimization

Fa' 2011 EE 382V: Dynamic Compilation

Students

Current PhD students

2025/– Chenyu Wang

2024/– Zander Ingare

2023/– Jeffrey Ma

2023/– Oishii Banerjee

2022/– Ikechukwu Uchendu

2022/– Jason Jabbour

2022/– Jason Yik

2022/– Jessica Quaye

2021/– Emma Chen

2021/– Mark Mazumder

2020/– Shvetank Prakash

2019/– Radhika Ghosal

Graduated PhD students

2019–2024 **Max Lam**
PhD Thesis: "Systems and Algorithms for Efficient, Secure and Private Machine Learning Inference,"
First Job: Research Scientist, Apple.

2019–2024 **Colby Banbury**
PhD Thesis: "Efficient and Scalable Tiny Machine Learning,"
First Job: Senior Research Scientist, Microsoft Research.

2019–2024 **Yu-shun Hsiao**
PhD Thesis: "Safety-Aware System Optimization for Autonomous Machines,"
First Job: CEO, Founder, Robotics Start-up.

2018–2024 **Srivatsan Krishnan**
PhD Thesis: "Designing Efficient Domain-Specific Architectures for Autonomous Systems,"
First Job: Senior Software Engineer, NVIDIA.

2018–2022 **Brian Plancher**
PhD Thesis: "GPU Acceleration for Real-time, Whole-body, Nonlinear Model Predictive Control,"
First Job: Assistant Professor at Barnard College at University of Columbia.

2014–2022 **Behzad Boroujerdian**
PhD Thesis: "Agile Development of Domain-Specific Solutions for Emerging Mobile Systems,"
First Job: Deep Learning Researcher, NVIDIA.

2014–2022 **Daniel Richins**
PhD Thesis: "Bottlenecks in Big PhD Thesis: Data Analytics and AI Applications and Opportunities for Improvement,"
First Job: Instructor, Brigham Young University.

2013–2018 **Yazhou Zu**
PhD Thesis: "Active Timing Margin Management to Improve Microprocessor Power Efficiency,"
First Job: Software Engineer, Google.

2011–2016 **Jingwen Leng**
PhD Thesis: "Guardband Management in Heterogeneous Architectures,"
First Job: Assistant Professor at Shanghai Jiao Tong University (CSE).

2011–2016 **Yuhao Zhu**
PhD Thesis: "Energy-Efficient Mobile Web Computing,"
First Job: Assistant Professor at Univ. of Rochester (CS).

Postdoctoral Associates

2020–2024 Matthew Stewart

2020–2023 Sabrina Neuman, Assistant Professor at Boston University.

M.S. students

2019–2021 Jonathan Cruz, Harvard

2015–2017 Wenzhi Cui, UT Austin

2014–2017 Matthew Halpern, UT Austin

2011–2013 Aditya Srikanth, UT Austin

2011–2013 Ankita Garg, UT Austin

258 publications · 107 conference · 51 journal · 94 arXiv · 4 workshop

Bio

Vijay Janapa Reddi is the Gordon McKay Professor of Electrical & Computer Engineering at Harvard University, Vice President and co-founder of MLCommons (mlcommons.org), and (from July 2026) Visiting Professor at ETH Zürich. As Vice President of MLCommons, he oversees the MLCommons Research organization and serves on the Board of Directors. He co-led the development of the MLPerf Inference benchmark (now the industry standard for ML benchmarking across datacenter, edge, mobile, and IoT, adopted by Google, Microsoft, NVIDIA, Meta, AMD, and Intel) and the MLPerf Power benchmark for sustainable AI. Before joining Harvard, he was an Associate Professor at the University of Texas at Austin's Department of Electrical and Computer Engineering. Drawing on his expertise in computer architecture, runtime systems, and applied machine learning, he creates innovative solutions at the intersection of mobile, edge, and embedded intelligence.

Dr. Janapa Reddi has earned numerous awards, including the Gilbreth Lectureship Honor from the National Academy of Engineering (NAE) in 2016, the IEEE TCCA Young Computer Architect Award (2016), and the Intel Early Career Award (2013). His research has been recognized with Best Paper Awards at the 2024 Vail Computer Elements Workshop (VCEW), the 2020 Design Automation Conference (DAC), the 2009 International Symposium on High-Performance Computer Architecture (HPCA), and the 2005 International Symposium on Microarchitecture (MICRO). His papers have been selected for IEEE Micro Top Picks in Computer Architecture in 2026, 2025, 2021, 2017, 2011, and 2010, with honorable mentions in 2023, 2022, and 2016, and the MLPerf Inference benchmark was selected for inclusion in the ISCA@50 25-Year Retrospective. He is a recipient of multiple Google Faculty Research Awards (2012, 2013, 2015, 2017, and 2020), the ACM SIGPLAN Programming Languages Software Award (2020), and the BenchCouncil Rising Star Award (2021). He is inducted into the MICRO Hall of Fame (2018) and the HPCA Hall of Fame (2019).

Dr. Janapa Reddi is strongly devoted to expanding access to applied machine learning for STEM, diversity, and the application of AI for social good. He is the founder and lead author of the open-source *Machine Learning Systems* textbook and curriculum ecosystem (mlsysbook.ai), used at 50+ universities across five continents, with two MIT Press hardcover volumes forthcoming in 2026 and 2027. Through the TinyML4D Academic Network (with ICTP), he ships hardware kits and curriculum to universities across Latin America, Africa, and Asia. He also founded and led the TinyML Professional Certificate on HarvardX/edX in partnership with Google and the tinyML Foundation, reaching 100,000+ learners worldwide and recognized as one of ClassCentral's 100 Most Popular Free Online Courses (2021) and the CogX Best Course in AI award (2021). At UT Austin, he previously led the Hands-on Computer Science (HaCS) program for K–12 students in partnership with the Austin Independent School District.

Dr. Janapa Reddi holds degrees in computer science from Harvard University (Ph.D.), computer engineering from the University of Colorado Boulder (M.S.), and computer engineering from Santa Clara University (B.S.). His passion is helping individuals and teams succeed while making the world a better place.

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